Effective Nuclear Charge and Shielding: Worksheet

Name _____ ID _____ Period ____ Date _____ Score _____ / 30

Use Z_eff = Z − core, where core is every electron below the outer ring. Pull = Z_eff ÷ n^2, where n is the ring number of the outer electron. The pull is a classroom model number.

Part 1. Estimate (1 point each, 6 points)

1. How many core electrons does nitrogen (Z = 7) have? _____
2. Estimate Z_eff for magnesium (Z = 12). _____
3. Estimate Z_eff for chlorine (Z = 17). _____
4. How many core electrons does potassium (Z = 19) have? _____
5. Which ring number holds the outer electron of boron (Z = 5)? _____
6. Estimate Z_eff for sulfur (Z = 16). _____

Part 2. Compare (2 points each, 8 points)

Circle the atom that holds its outer electron harder. Show Z_eff and n for both atoms.

7. lithium (Z = 3) or potassium (Z = 19) They hold it equally? Circle one of the three.
   Z_eff and n for each: _____
8. sodium (Z = 11) or potassium (Z = 19) They hold it equally? Circle one of the three.
   Z_eff and n for each: _____
9. magnesium (Z = 12) or chlorine (Z = 17) They hold it equally? Circle one of the three.
   Z_eff and n for each: _____
10. sodium (Z = 11) or sulfur (Z = 16) They hold it equally? Circle one of the three.
    Z_eff and n for each: _____

Scoring, items 7 to 10 (each)	Points
Circled the atom that holds it harder	1
Z_eff and n shown for both atoms, matching the choice	1

Part 3. Picture (4 points)

11. Draw a sodium atom (Z = 11) on the blank space below: the nucleus with its charge, and one ring for each group of electrons. Shade the core rings. Write Z_eff next to your drawing.

Scoring, item 11	Points
Nucleus with its charge written	1
Rings drawn with the right electron count in each	1
Core rings shaded and the outer ring left unshaded	1
Z_eff written and matched to the drawing	1

Part 4. Across period 3 (4 points)

12. Fill in Z_eff for each atom.

Na (11)	Mg (12)	Al (13)	Si (14)	P (15)	S (16)	Cl (17)	Ar (18)

Scoring, item 12	**Points**
All eight boxes correct	4
Six or seven correct	3
Four or five correct	2
Two or three correct	1
Fewer than two correct	0

Part 5. Explain (6 points)

13. Potassium has more protons than sodium, yet its outer electron comes off more easily. Explain, using the words core, Z_eff and ring.

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Scoring, item 13	**Points**
Claim: says which atom holds its outer electron harder and which is easier to remove	2
Evidence: gives Z_eff and the ring number for both atoms	2
Reasoning: connects equal Z_eff and the farther ring to the weaker pull	2

Part 6. Predict (2 points)

14. Moving from sodium to chlorine, what does the rising pull on the outer electron predict about the size of the atom? Say what you expect and why.

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Scoring, item 14	**Points**
States the change in size	1
Gives a reason that uses the pull on the outer electrons	1